In [1]:
```
import xarray as xr
import cartopy.crs as ccrs
import matplotlib.pyplot as plt
import cartopy.feature as cf
from cartopy.util import add_cyclic_point
```

In [2]:
```
# Imports

echam_ds = xr.load_dataset('samples/HW310_esm-flat10_RESCUE_echam6_echam_186
tracer_ds = xr.load_dataset('samples/HW310_esm-flat10_RESCUE_echam6_tracer_1
tracerpl_ds = xr.load_dataset('samples/HW310_esm-flat10_RESCUE_echam6_tracer
co2_surface_ds = xr.load_dataset('samples/HW310_esm-flat10_RESCUE_echam6_co2
```

In [3]:
```
echam_ds
```

Out[3]: xarray.Dataset

► **Dimensions:** (nhyi: 96, nhym: 95, **time**: 12, **lev**: 95, **lat**: 96, **lon**: 192, nsp: 2080, nc2: 2)

▼ **Coordinates:**

time	(time)	datetime64[ns]	1862-01-31T23:52:00 ... 1862-12-...	
lev	(lev)	float64	1.0 2.0 3.0 4.0 ... 93.0 94.0 95.0	
lat	(lat)	float64	88.57 86.72 84.86 ... -86.72 -88.57	
lon	(lon)	float64	0.0 1.875 3.75 ... 356.2 358.1	

► **Data variables:** (147)

▼ **Attributes:**

CDI :	Climate Data Interface version 2.4.2 (https://mpimet.mpg.de/cdi)
Conventions :	CF-1.6
institution :	Max Planck Institute for Meteorology
history :	Mon Jul 06 14:14:01 2026: cdo -O -r -f nc4c -z zip_3 -t /gxfs_work/geomar/smomw271/FOCI/flat10/echam.codes copy /gxfs_work/geomar/smomw271/FOCI/flat10/HW310_esm-flat10_RESCUE/echam/HW310_esm-flat10_RESCUE_echam6_echam_1862.grb /gxfs_work/geomar/smomw271/FOCI/flat10/HW310_esm-flat10_RESCUE/echam/HW310_esm-flat10_RESCUE_echam6_echam_1862.nc
CDO :	Climate Data Operators version 2.4.2 (https://mpimet.mpg.de/cdo)

In [4]:
```
# slice dataset
# time format '1862-01-31T23:52:00.000000000'
time_slice = echam_ds.sel(time=slice('1862-01-31T23:52:00.000000000', '1862-
echam_ds.time

lon_slice = echam_ds.sel(lon=slice(10, 15))
lon_slice
```

```
# extract variables
echam_ds.sel(lev=1).q  # specific humidity [kg/kg]
echam_ds.sel(time='1862-01-31T23:52:00.000000000').st  # temperature [K]
echam_ds.sel(lat=slice(-25.0, -27.0)).tpot # potential temperature [K]
echam_ds.sel(lat=slice(27, 25)).aclcac # cloud cover
echam_ds.sel(nsp=137) # not sure what nsp is
```

Out[4]: xarray.Dataset

► **Dimensions:** (nhyi: 96, nhym: 95, **time**: 12, **lev**: 95, **lat**: 96, **lon**: 192, nc2: 2)

▼ **Coordinates:**

time	(time)	datetime64[ns]	1862-01-31T23:52:00 ... 1862-12-...
lev	(lev)	float64	1.0 2.0 3.0 4.0 ... 93.0 94.0 95.0
lat	(lat)	float64	88.57 86.72 84.86 ... -86.72 -88.57
lon	(lon)	float64	0.0 1.875 3.75 ... 356.2 358.1

► **Data variables:** (147)

▼ **Attributes:**

CDI :	Climate Data Interface version 2.4.2 (https://mpimet.mpg.de/cdi)
Conventions :	CF-1.6
institution :	Max Planck Institute for Meteorology
history :	Mon Jul 06 14:14:01 2026: cdo -O -r -f nc4c -z zip_3 -t /gxfs_work/geomar/smomw271/FOCI/flat10/echam.codes copy /gxfs_work/geomar/smomw271/FOCI/flat10/HW310_esm-flat10_RESCUE/echam/HW310_esm-flat10_RESCUE_echam6_echam_1862.grb /gxfs_work/geomar/smomw271/FOCI/flat10/HW310_esm-flat10_RESCUE/echam/HW310_esm-flat10_RESCUE_echam6_echam_1862.nc
CDO :	Climate Data Operators version 2.4.2 (https://mpimet.mpg.de/cdo)

In [5]:
```
tracer_ds # contains CO2 and levels 1-95
```

Out[5]: xarray.Dataset

► **Dimensions:** (nhyi: 96, nhym: 95, **time**: 12, **lev**: 95, **lat**: 96, **lon**: 192)

▼ **Coordinates:**

time	(time)	datetime64[ns]	1926-01-31T23:52:00 .....
lev	(lev)	float64	1.0 2.0 3.0 4.0 ... 93.0 9...
lat	(lat)	float64	88.57 86.72 84.86 ... -8...
lon	(lon)	float64	0.0 1.875 3.75 ... 356.2 ...

▼ **Data variables:**

hyai	(nhyi)	float64	0.0 1.989 2.693 ... 0.0 0...
hybi	(nhyi)	float64	0.0 0.0 0.0 ... 0.973 0.9...
hyam	(nhym)	float64	0.9946 2.341 3.119 ... 0....
hybm	(nhym)	float64	0.0 0.0 0.0 ... 0.9826 0....
CO2	(time, lev, lat, lon)	float32	0.0006916 0.0006916 ....

▼ **Attributes:**

CDI :	Climate Data Interface version 2.2.4 (https://mpimet.mpg.de/cdi)
Conventions :	CF-1.6
institution :	Max Planck Institute for Meteorology
history :	Sun Dec 28 18:18:19 2025: cdo -O -r -f nc4c -z zip_3 -t /user/haowei.wey02/u19858/esm/esm-experiments-mdc//HW310_esm-flat10_RESCUE/log/echam/HW310_esm-flat10_RESCUE_echam6_tracer.codes copy HW310_esm-flat10_RESCUE_echam6_tracer_1926.grb HW310_esm-flat10_RESCUE_echam6_tracer_1926.nc
CDO :	Climate Data Operators version 2.2.2 (https://mpimet.mpg.de/cdo)

In [6]:
```
tracerpl_ds # contains CO2 and pressure levels (plev) in Pa
# also geosp = surface geopotential (orography) [m**2/s**2] and aps surface
tracerpl_ds.CO2
```

Out[6]: xarray.DataArray 'CO2' (**time**: 12, **plev**: 26, **lat**: 96, **lon**: 192)

array([[[[0.00077713, 0.00077713, 0.00077713, ..., 0.00077713,
0.00077713, 0.00077713],
[nan, nan, nan, ..., 0.00077575,
0.00077579, nan],
[0.00077509, 0.00077505, 0.000775 , ..., 0.00077505,
0.00077507, 0.00077508],
...,
[nan, nan, nan, ..., nan,
nan, nan],
[nan, nan, nan, ..., nan,
nan, nan],
[nan, nan, nan, ..., nan,
nan, nan]],

[[0.00077425, 0.00077425, 0.00077425, ..., 0.00077425,
0.00077425, 0.00077425],
[0.00077366, 0.00077368, 0.0007737 , ..., 0.00077365,
0.00077365, 0.00077365],
[0.00077386, 0.00077388, 0.0007739 , ..., 0.00077389,
0.00077386, 0.00077385],
...
0.00075527, 0.00075527],
[0.00075529, 0.00075529, 0.00075529, ..., 0.00075528,
0.00075528, 0.00075529],
[0.00075529, 0.00075529, 0.00075529, ..., 0.00075529,
0.00075529, 0.00075529]],

[[0.0007546 , 0.0007546 , 0.0007546 , ..., 0.0007546 ,
0.0007546 , 0.0007546],
[0.00075455, 0.00075455, 0.00075456, ..., 0.00075455,
0.00075455, 0.00075455],
[0.00075452, 0.00075452, 0.00075452, ..., 0.00075452,
0.00075452, 0.00075452],
...,
[0.00075518, 0.00075518, 0.00075518, ..., 0.00075518,
0.00075518, 0.00075518],
[0.0007552 , 0.0007552 , 0.0007552 , ..., 0.0007552 ,
0.0007552 , 0.0007552],
[0.0007552 , 0.0007552 , 0.0007552 , ..., 0.0007552 ,
0.0007552 , 0.0007552]]]],
shape=(12, 26, 96, 192), dtype=float32)

▼ Coordinates:

time	(time)	datetime64[ns]	1946-01-31T23:52:00 ... 1946-12-...
plev	(plev)	float64	1e+05 9.25e+04 ... 20.0 10.0
lat	(lat)	float64	88.57 86.72 84.86 ... -86.72 -88.57
lon	(lon)	float64	0.0 1.875 3.75 ... 356.2 358.1

▼ Attributes:

long_name :	mass fraction of CO2 in air
units :	kg kg-1
code :	1
table :	255
CDI_grid_type :	gaussian
CDI_grid_num_L...	48

In [7]:
```
co2_surface_ds
```

Out[7]: xarray.Dataset

► Dimensions: (**time**: 12, **lat**: 96, **lon**: 192)

▼ Coordinates:

time	(time)	datetime64[ns]	1859-01-15 ... 1859-12-15
lat	(lat)	float64	88.57 86.72 84.86 ... -86.72 -88.57
lon	(lon)	float64	0.0 1.875 3.75 ... 356.2 358.1

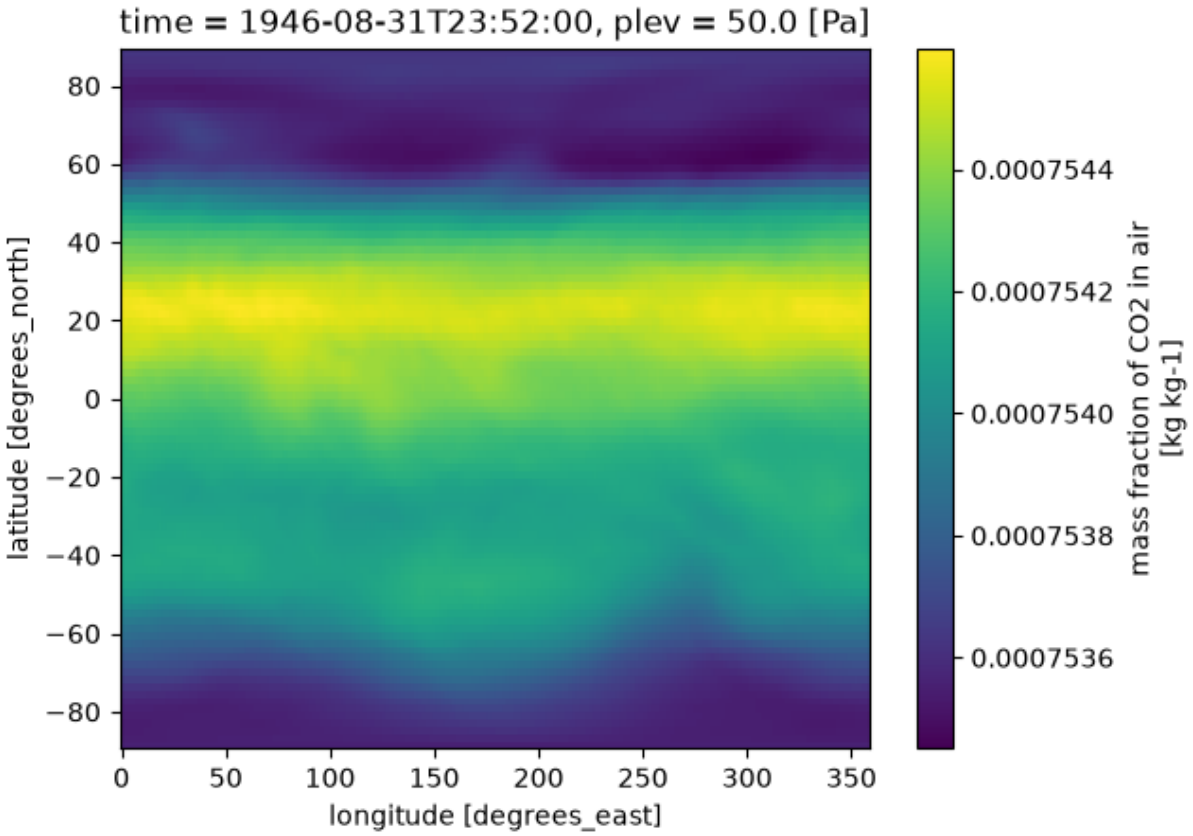
► Data variables: (25)

▼ Attributes:

CDI :	Climate Data Interface version 2.2.4 (https://mpimet.mpg.de/cdi)
Conventions :	CF-1.6
source :	ECHAM6
institution :	Max Planck Institute for Meteorology
history :	Fri Dec 19 12:49:52 2025: cdo -O -t echam6 -r -f nc4c -z zip_1 -settunits,days -settaxis,1859-01-15,00:00:00,1months HW310_esm-flat10_RESCUE_echam6_co2_mm_1859.grb HW310_esm-flat10_RESCUE_echam6_co2_mm_1859.nc
CDO :	Climate Data Operators version 2.2.2 (https://mpimet.mpg.de/cdo)

In [8]:
```
# Plot CO2 data
data = tracerpl_ds.sel(plev=50, time='1946-08-31T23:52:00.000000000')
data["CO2"].plot()
```

Out[8]: <matplotlib.collections.QuadMesh at 0x76a59d396cf0>



In [9]:

```
## Setup Cartopy
projection = ccrs.Robinson(central_longitude=180)

# Specify CRS, that will be used to tell the code, where should our data be
crs = ccrs.PlateCarree()

# Now we will create axes object having specific projection
plt.figure(dpi=150)
ax = plt.axes(projection=projection, frameon=True)

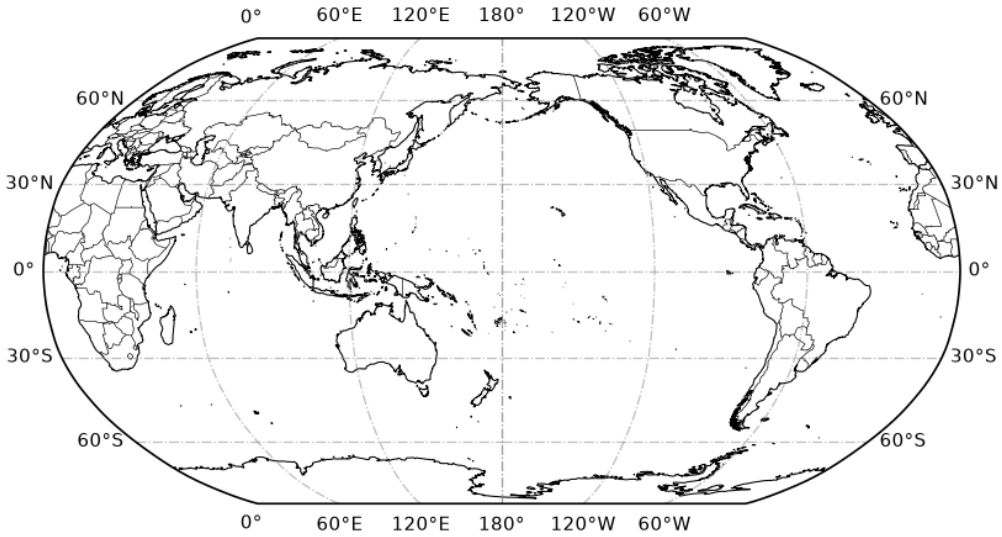
# Draw gridlines in degrees over Mercator map
gl = ax.gridlines(crs=crs, draw_labels=True,
                  linewidth=.6, color='gray', alpha=0.5, linestyle='-.')
gl.xlabel_style = {"size" : 7}
gl.ylabel_style = {"size" : 7}

# To plot borders and coastlines, we can use cartopy feature
ax.add_feature(cf.COASTLINE.with_scale("50m"), lw=0.5)
ax.add_feature(cf.BORDERS.with_scale("50m"), lw=0.3)

# Now, we will specify extent of our map in minimum/maximum longitude/latitu
# Note that these values are specified in degrees of longitude and degrees o
# However, we can specify them in any crs that we want, but we need to provi
# crs argument in ax.set_extent

# lon_min = -20
# lon_max = 45
# lat_min = 34
# lat_max = 60
```

```
## crs is PlateCarree -> we are explicitly telling axes, that we are creatin
# ax.set_extent([lon_min, lon_max, lat_min, lat_max], crs=crs)
plt.title(f"Sample Global Map")
plt.show()
```



In [10]:
```
def wrap_data(data, key):
    da = data[key]
    lon_idx = list(da.dims).index('lon')
    cyclic_data, cyclic_lons = add_cyclic_point(da.values, coord=da.lon.valu
    new_coords = dict(da.coords)
    new_coords['lon'] = cyclic_lons

    cyclic_da = xr.DataArray(
        cyclic_data,
        coords=new_coords,
        dims=da.dims,
        attrs=da.attrs
    )
    return cyclic_da

def map_setup():
    projection = ccrs.Robinson()
    crs = ccrs.PlateCarree()
    plt.figure(dpi=150)
    ax = plt.axes(projection=projection, frameon=True)
    gl = ax.gridlines(crs=crs, draw_labels=True,
                      linewidth=.6, color='gray', alpha=0.5, linestyle='-.')
    gl.xlabel_style = {"size" : 7}
```

```
    gl.ylabel_style = {"size" : 7}
    ax.add_feature(cf.COASTLINE.with_scale("50m"), lw=0.5)
    # ax.add_feature(cf.BORDERS.with_scale("50m"), lw=0.3)
    return ax


def plot_dataset(data, key):
    # data repair
    data = wrap_data(data, key)
    ax = map_setup()

    # plot data
    cbar_kwargs = {'orientation':'horizontal', 'shrink':1, "pad":.05, 'aspec
    contour_plot = data.plot.contourf(ax=ax, transform=ccrs.PlateCarree(), c
    contour_plot.colorbar.ax.tick_params(labelsize="small")

    # ax.set_extent([lon_min, lon_max, lat_min, lat_max], crs=crs)
    plt.title(data.long_name)
    plt.show()

sliced = tracerpl_ds.sel(plev=20, time='1946-08-31T23:52:00.000000000')
plot_dataset(sliced, "CO2")
```

